

Unit 3

Set and Functions

Sr. No.	Questions	A	B	C	D
1	The set builder form of the set $\{1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots\}$ is:	$\{x x = \frac{1}{n}, n \in W\}$	$\{x x = \frac{1}{2n+1}, n \in W\}$ ✓	$\{x x = \frac{1}{n+1}, n \in W\}$	$\{x x = 2n+1, n \in W\}$
2	If $A = \{\}$, then $P(A)$ is:	$\{\}$	$\{1\}$	$\{\{\}\}$ ✓	$\varnothing$
3	If $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $U - (A \cap B)$ is:	$\{1, 2, 4, 5\}$ ✓	$\{2, 3\}$	$\{1, 3, 4, 5\}$	$\{1, 2, 3\}$
4	If A and B are overlapping sets, then $n(A \cap B)$ is equal to:	$n(A)$	$n(B)$	$A \cap B$	$n(A) - n(A \cap B)$ ✓
5	If $A \subseteq B$ and $B - A \neq \varnothing$, then $n(B - A)$ is equal to:	0	$n(B)$	$n(A)$	$n(B) - n(A)$ ✓
6	If $n(A \cup B) = 50$, $n(A) = 30$, and $n(B) = 35$, then $n(A \cap B) =$:	23	15✓	9	40
7	If $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$, then the Cartesian product of A and B contains exactly _____ elements.	13	12✓	10	6
8	If $f(x) = x^2 - 3x + 2$, then the value of $f(a+1)$ is equal to:	$a+1$	a^2+1	a^2+2a+1	a^2-a ✓
9	Given that $f(x) = 3x + 1$, if $f(x) = 28$, then the value of x is:	9✓	27	3	18
10	Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$; two non-empty sets and $f: A \rightarrow B$ be a function defined as $f = \{(1, a), (2, a), (3, b)\}$, then which of the following statements is true?	f is injective	f is surjective✓	f is bijective	f is into only

Solution of MCQs

1	Elements are odd reciprocals $\Rightarrow \frac{1}{2n+1}$
2	Power set of empty set = $\{\{\}\}$ or $\{\varnothing\}$
3	$A \cap B = \{3\}$, $U - A \cap B = \{1, 2, 4, 5\}$

4	$n(A \cap B) = n(A) - n(A \cap B)$ $\text{Let } A = \{1, 2\}$ $\text{and } B = \{1\}$ $A \cap B = \{1\}$ $\text{Now } n(A \cap B) = n(A) - n(A \cap B)$ $1 = 2 - 1$ $1 = 1$
5	$B - A = \text{elements only in } B$ $\Rightarrow n(B) - n(A)$
6	$n(A \cup B) = 50, \quad n(A) = 30, \quad \text{and} \quad n(B) = 35, \quad \text{then} \quad n(A \cap B) = ?$ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ $50 = 30 + 35 - n(A \cap B)$ $n(A \cap B) = 30 + 35 - 50$ $n(A \cap B) = 15$
7	$4 \text{ elements} \times 3 \text{ elements} = 12$
8	$f(a + 1) = ?$ $f(x) = x^2 - 3x + 2$ $f(a + 1) = (a + 1)^2 - 3(a + 1) + 2$ $f(a + 1) = (a)^2 + 2(a)(1) + (1)^2 - 3a - 3 + 2$ $f(a + 1) = a^2 + 2a + 1 - 3a - 3 + 2$ $f(a + 1) = a^2 - a$
9	$f(x) = 3x + 1 = 28 \Rightarrow 3x = 28 - 1 \Rightarrow x = 9$
10	Every element in B is mapped, so f is surjective.

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